ASSIGNMENT No: 03 (BI)
Name: Aditya Rade
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Title: Protein Structure Prediction
Code:

```
protein_sequence="MKTIIALSYIFCLVFADYKDDDDK"
protein_sequence=protein_sequence.upper()

print("PROTEIN STRUCTURE PREDICTION")
print("-"*50)
print("Protein Sequence:")
print(protein_sequence)
print("\nProtein Length:",len(protein_sequence),"amino acids")

print("\n1. AMINO ACID COMPOSITION")
print("-"*50)

amino_acids="ACDEFGHIKLMNPQRSTVWY"

for aa in amino_acids:
   count=protein_sequence.count(aa)
   if count>0:
      print(aa,":",count)

print("\n2. BASIC PROTEIN PROPERTIES")
print("-"*50)

hydrophobic="AILMFWVY"
charged="DEKR"

hydrophobic_count=sum(protein_sequence.count(aa) for aa in hydrophobic)
charged_count=sum(protein_sequence.count(aa) for aa in charged)

print("Hydrophobic amino acids:",hydrophobic_count)
print("Charged amino acids:",charged_count)

print("\n3. FASTA FORMAT")
print("-"*50)

fasta=">Target_Protein\n"+protein_sequence
print(fasta)

with open("target_protein.fasta","w") as file:
   file.write(fasta)

print("\nFASTA file created successfully.")
```

```
print("File name: target_protein.fasta")

print("\n4. HOMOLOGY MODELING WORKFLOW")
print("-"*50)

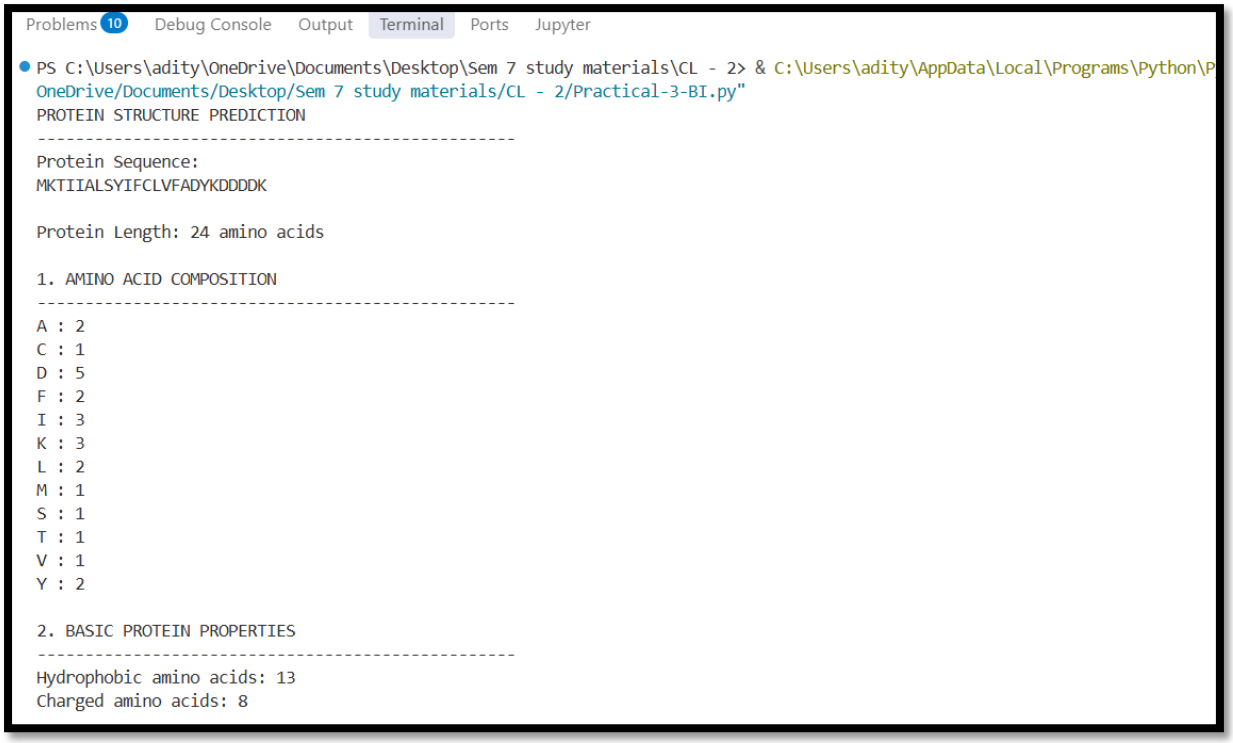
print("Step 1: Submit the protein sequence to a")
print("       sequence similarity/search tool.")
print("Step 2: Identify homologous proteins with")
print("       experimentally determined structures.")
print("Step 3: Select the best template based on")
print("       sequence identity and coverage.")
print("Step 4: Align target and template sequences.")
print("Step 5: Build the 3D model.")
print("Step 6: Refine and validate the model.")

print("\nRESULT")
print("-"*50)

print("Target protein sequence analyzed successfully.")
print("FASTA file generated successfully.")
print("The sequence can now be used for homology")
print("modeling and 3D structure prediction.")
print("Aditya Rade   BEAD23167")
```

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OUTPUT:



```
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2> & C:\Users\adity\AppData\Local\Pro
OneDrive/Documents/Desktop/Sem 7 study materials/CL - 2/Practical-3-BI.py"

3. FASTA FORMAT
--------------------------------------------------
>Target_Protein
MKTIIALSYIFCLVFADYKDDDDK

FASTA file created successfully.
File name: target_protein.fasta

4. HOMOLOGY MODELING WORKFLOW
--------------------------------------------------
Step 1: Submit the protein sequence to a
        sequence similarity/search tool.
Step 2: Identify homologous proteins with
        experimentally determined structures.
Step 3: Select the best template based on
        sequence identity and coverage.
Step 4: Align target and template sequences.
Step 5: Build the 3D model.
Step 6: Refine and validate the model.

RESULT
--------------------------------------------------
Target protein sequence analyzed successfully.
FASTA file generated successfully.
The sequence can now be used for homology
modeling and 3D structure prediction.
Aditya Rade   BEAD23167
PS C:\Users\adity\OneDrive\Documents\Desktop\Sem 7 study materials\CL - 2>
```